



# HostelHub

## All-in-One Digital Hostel Management Platform

Streamlining Maintenance Complaints, Daily Roll Call Attendance, Mess Regulations & Real-Time Batch Group Communication

Developed by Abhinav Kumar (Lead Full-Stack Developer) • August 2026



## ⚠️ Traditional Hostel Pain Points

- **Manual Register Attendance:** High proxy error rates and physical record loss.
- **Unorganized Repair Dispatch:** Long delays in electrician and plumber assignment.
- **Food Waste in Mess:** No prior tracking of meal skips causing excess cooking.
- **Unverified Communication:** Fragmented chat groups with zero warden moderation.
- **No Proof Accountability:** Service staff resolving tickets without proof photo upload.

## 🎯 The HostelHub Solution

- **100% Digital Workflow:** Unifies Students, Wardens, Staff & Admins in real-time.
- **Automated Ticket Dispatching:** Category-based routing with resolution SLA timeline.
- **Mandatory Resolution Proof:** Staff uploads job completion photo before closing ticket.
- **Hostel Block Roll Call:** Instant filterable statistics (Total, Present, Absent).
- **Evening Snacks & Skip System:** Cuts mess food waste by up to 30%.



## Core Modular Pillars

### Maintenance Ticket Engine:

Camera photo capture, priority tags, staff assignment & star ratings.

### Daily Roll Call Roster:

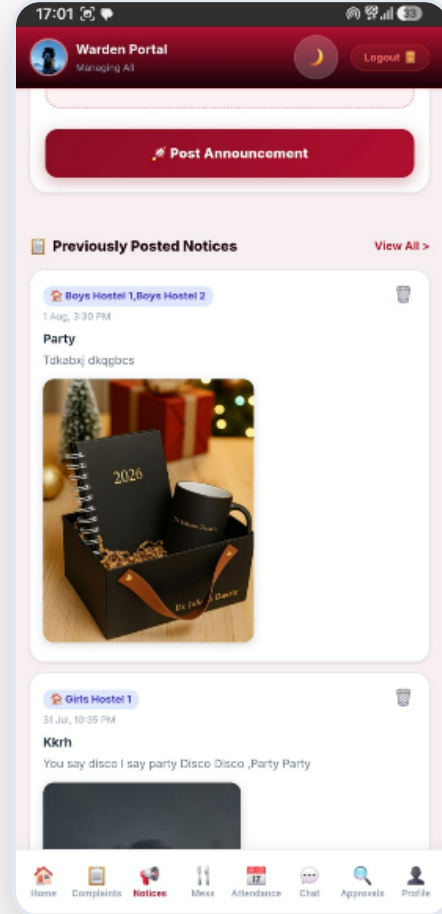
Binary Present/Absent marking with hostel block filtering.

### Mess Regulations & Snacks:

Weekly menu schedule, evening snacks item & meal skip tracking.

### Real-Time Group Chat:

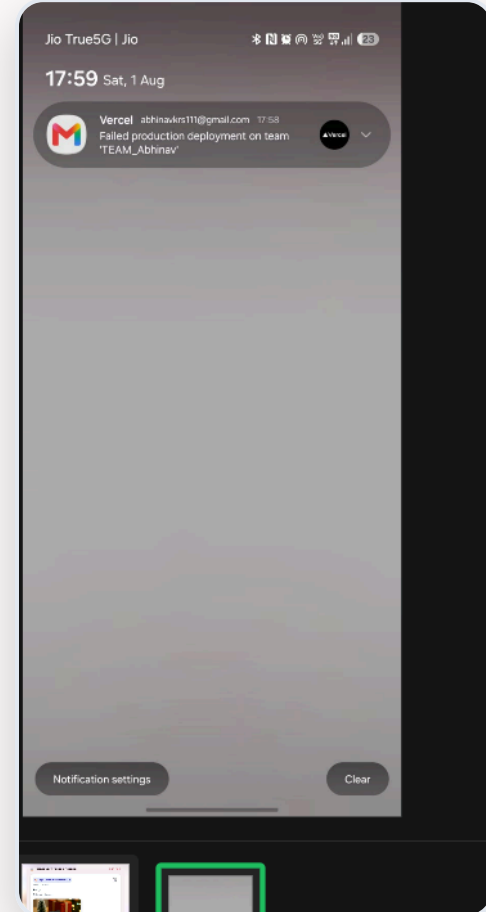
Block & Batch chats powered by Socket.io WebSockets.





### Key Student Capabilities

- **Categorized Issue Selection:** Electrical, Plumbing, Carpentry, Cleaning, Other.
- **Urgency Priority Tags:** Low, Medium, High, and Urgent severity tags.
- **Native Photo Proof Capture:** Attach photo evidence directly using phone camera.
- **4-Stage Ticket Progress Timeline:** Visual progression tracking (Ticket Raised → Staff Assignment → Work Progress → Job Resolved).
- **Direct Warden Contact Cards:** One-tap phone/email buttons to contact assigned block wardens.





### 1. Raise Ticket

Student submits issue title, description, category, priority tag & issue photo attachment.

### 2. Assign Staff

Warden reviews new complaints & assigns available electrician, plumber or technician.

### 3. Work Progress

Staff marks job 'In Progress', inspects site & conducts physical maintenance repair.

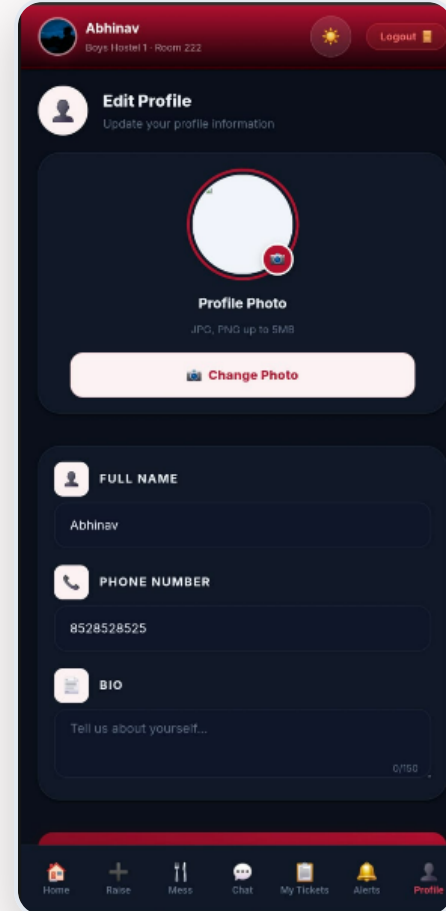
### 4. Proof & Feedback

Staff uploads completion photo proof. Student inspects work & rates 1-5 stars.



## Real-Time Attendance Management

- **Strict Binary Status Model:** Tracks Present & Absent (Outing clutter eliminated).
- **Hostel Block Filter:** Filter by All Hostels, Boys Hostel 1/2, or Girls Hostel 1/2.
- **Summary Metrics Bar:** High-contrast cards for Total Students, Present Count & Absent Count.
- **Bulk Actions & Remarks:** One-tap 'Mark All Present' with custom student remarks.
- **Student Attendance History:** Students monitor personal attendance % & history logs.





## 4-Meal Schedule

- **Breakfast:** Morning nutritious meals & beverages.
- **Lunch:** Full noon meal menu items.
- **Evening Snacks (  Added):** Evening tea & snacks refreshment.
- **Dinner:** Evening meal & special weekend menus.
- **Warden Menu Controls:** Real-time menu updates by Warden.

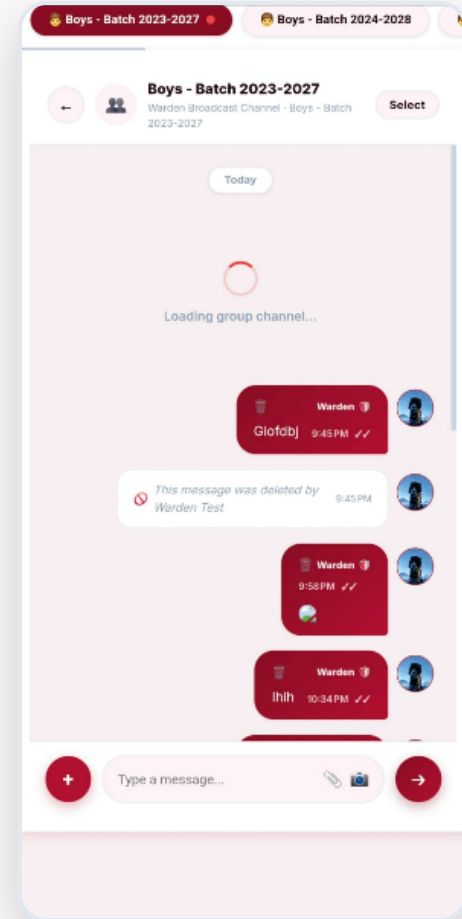
## Meal Skip & Waste Prevention

- **Student Skip Toggle:** One-tap toggle for upcoming meals.
- **Mess Kitchen Analytics:** Real-time headcount of expected diners.
- **Cuts Food Waste:** Reduces food waste by up to 30%.
- **Mess Feedback System:** Ratings on hygiene, quality & behavior.



## 💬 Live Socket.io Messaging & Support

- **Automatic Group Membership:** Joined into Block Chat & Batch Chat on signup.
- **Official Announcements:** Wardens broadcast notices to specific or all hostel blocks.
- **Image Attachment Sharing:** Share repair photos & academic notices.
- **Developer Team Cards:** Direct links to GitHub, LinkedIn, Twitter, Email & 📷 Instagram.







## Full Stack Technology

- **Frontend:** Angular v19 Standalone, TypeScript, RxJS, Vanilla CSS.
- **Mobile APK:** Capacitor Native Android Bridge & Back-Button Guard.
- **Backend API:** Node.js, Express.js REST Controllers & Auth Middlewares.
- **Database:** PostgreSQL with Sequelize ORM.
- **Cloud Storage:** Supabase Storage with Base64 fallback.

## Security Enforcements

- **Profile Re-Approval Rule:** Editing Gender or Batch triggers critical warning & requires Warden Re-approval.
- **Dual-Press Back Exit Guard:** Back button navigates to Home tab first; second press opens Exit Confirmation Modal.
- **JWT Role Protection:** Route guards for Student, Warden, Staff & Admin.

# Transforming Hostel Management with HostelHub

✓ Zero Paperwork • ✓ 100% Ticket Accountability • ✓ Reduced Mess Food Waste • ✓ Real-Time Safety

## Thank You!

For app deployment or technical inquiries, contact Abhinav Kumar (Lead Full-Stack Developer).